The Solutions to Exam 2026

B—bookwork, A—new application, E—new example, T—new theory

1.

a) We have the following outcomes:

$x_1\ x_2$	00	01	10	11
$y = \max(x_1, x_2)$	0	1	1	1
Prob.	0.09	0.21	0.21	0.49

i) Thus $P(y=0) = 0.09$, $P(y=1) = 0.91$,

$H(y) = -0.09 \log(0.09) - 0.91 \log(0.91) = 0.313 + 0.124 = 0.437$ bits [3E]

ii) We have the joint distribution

$x_1 \setminus y$	0	1
0	0.09	0.21
1	0	0.7

$H(y|x_1) = 0.3\ H(0.3) + 0.7\ H(1) = 0.264 + 0 = 0.264$ [3E]

$I(x_1\ ;\ y) = H(y) - H(y|\ x_1) = 0.439 - 0.264 = 0.175$ bits

iii) $I(x_{1:2}\ ;\ y) = H(y) - H(y|\ x_{1:2})$

$= 0.437 - 0$ y is a function of x_1 and x_2

$= 0.437$ bits [3E]

b) Recall

$$D(\boldsymbol{p}||\boldsymbol{q}) = \sum_i p_i \log_2\left(\frac{p_i}{q_i}\right) \geq 0$$

$D(\mathbf{p}||\mathbf{q}) = \frac{1}{2}\log 2 + \frac{1}{4}\log 1 + \frac{1}{8}\log\frac{1}{2} + \frac{1}{8}\log\frac{1}{2} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$ [4E]

$D(\mathbf{q}||\mathbf{p}) = \frac{1}{4}\log\frac{1}{2} + \frac{1}{4}\log 1 + \frac{1}{4}\log 2 + \frac{1}{4}\log 2 = -\frac{1}{4} + \frac{1}{2} = \frac{1}{4}$ [4E]

c) The probability of any given value of $x_{1:4}$ depends on the number of 1's and 0's. We create four subsets with equal probabilities to generate a pair of bits and two other subsets to generate one bit only. The expected number of bits generated is [4B]

$$EK = 8p(1-p)^3 + 10p^2(1-p)^2 + 8p^3(1-p)$$ [4B]

2. a) [1B each]

(1) total probability of the jointly typical set

(2) $p(x,y) \leq \max_{x,y \in J_\varepsilon^{(n)}} p(x,y)$

(3) $\max_{x,y \in J_\varepsilon^{(n)}} p(x,y) \leq 2^{-n(H(x,y)-\varepsilon)}$, from definition of the jointly typical set

(4) algebra

(5) total probability of the jointly typical set <= 1

(6) $p(\boldsymbol{x},\boldsymbol{y}) \geq \min_{\boldsymbol{x},\boldsymbol{y} \in J_\varepsilon^{(n)}} p(\boldsymbol{x},\boldsymbol{y})$

(7) $\min_{\boldsymbol{x},\boldsymbol{y} \in J_\varepsilon^{(n)}} p(\boldsymbol{x},\boldsymbol{y}) = 2^{-n(H(x,y)+\varepsilon)}$, from definition of the jointly typical set

(8) algebra

b)

i) Note that the marginal distribution of $\boldsymbol{x}$ is

$P(\boldsymbol{x}=0)=5/9, \quad P(\boldsymbol{x}=1)=4/9.$ [2E]

Because $\varepsilon = 0$, $x \in T_x$ if and only if exactly 5 of the $\boldsymbol{x}_i$'s are equal to 0, and the other 4 of the $\boldsymbol{x}_i$'s are equal to 1. Hence

$$P(x \in T_x) = \binom{9}{5}\left(\frac{5}{9}\right)^5\left(\frac{4}{9}\right)^4 = 0.261.$$ [3E]

ii) If, in addition, $(x,y) \in J_0^{(9)}$, we require that $\boldsymbol{y}_i = 0$ for 4 out of 5 index i's for which $\boldsymbol{x}_i = 0$, and $\boldsymbol{y}_i = 0$ for 2 out of 4 index i's for which $\boldsymbol{x}_i = 1$.
Thus

$$P(x,y \in J_0^{(9)} | x \in T_x) = \binom{5}{4}\left(\frac{4}{5}\right)^4\left(\frac{1}{4}\right)^1 \times \binom{4}{2}\left(\frac{2}{4}\right)^2\left(\frac{2}{4}\right)^2 = 0.887.$$ [3E]

iii) Hence we determine the value

$$\begin{aligned} P\left(x,y \in J_0^{(9)}\right) &= P\left(x,y \in J_0^{(9)} \middle| x \in T_x\right) P(x \in T_x) \\ &\quad + P\left(x,y \in J_0^{(9)} \middle| x \notin T_x\right) P(x \notin T_x) \\ &= P\left(x,y \in J_0^{(9)} \middle| x \in T_x\right) P(x \in T_x) + 0 = 0.887 \times 0.261 \\ &= 0.232 \end{aligned}$$

[3E]

iv) We also require that $\boldsymbol{z}_i = 0$ for 4 out of 5 index i's for which $\boldsymbol{x}_i = 0$, and $\boldsymbol{z}_i = 0$ for 2 out of 4 index i's for which $\boldsymbol{x}_i = 1$. But, since $\mathbf{z}$ is independent of $\mathbf{x}$,

$$P(x,z \in J_0^{(9)} | x \in T_x) = \binom{5}{4}\left(\frac{2}{3}\right)^4\left(\frac{1}{3}\right)^1 \times \binom{4}{2}\left(\frac{2}{3}\right)^2\left(\frac{1}{3}\right)^2 = 0.097$$ [3E]

Thus

$$\begin{aligned} P\left(x,z \in J_0^{(9)}\right) &= P\left(x,z \in J_0^{(9)} \middle| x \in T_x\right) P(x \in T_x) \\ &= 0.097 \times 0.261 = 0.025 \end{aligned}$$

[3E]

3.

a)

000000110101001000111100

Parsing: 0, 00, 000, 1, 10, 101, 001, 0001, 11, 100 [5E]

Locations: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

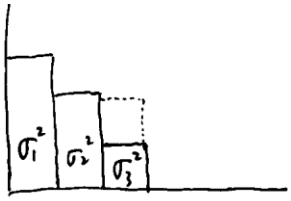
There are 10 phrases, so we need 4 bits to represent the locations.

Encoding: (0000, 0), (0001, 0), (0010, 0), (0000, 1), (0100, 0) [5E]

(0101, 1), (0010, 1), (0011, 1), (0100, 1), (0101, 0)

b)

i)



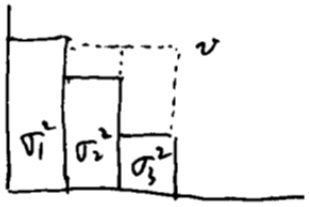
Single channel is when

$$3P \leq \sigma_2^2 - \sigma_3^2 \quad \text{[1E]}$$

Capacity

$$C = \frac{1}{2}\log\left(1 + \frac{3P}{\sigma_i^2}\right) \quad \text{[2E]}$$

ii)



A pair of channel is when

$$\sigma_2^2 - \sigma_3^2 < 3P \leq \sigma_1^2 - \sigma_2^2 + \sigma_1^2 - \sigma_3^2$$

$$= 2\sigma_1^2 - \sigma_2^2 - \sigma_3^2 \qquad \text{[2E]}$$

$$3P = \nu - \sigma_2^2 + \nu - \sigma_3^2 \implies \nu = \frac{3P + \sigma_2^2 + \sigma_3^2}{2}$$

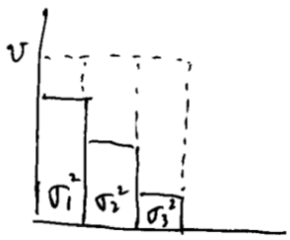
$$P_2 = \nu - \sigma_2^2 = \frac{3P - \sigma_2^2 + \sigma_3^2}{2} \qquad \text{[2E]}$$

$$P_3 = \nu - \sigma_3^2 = \frac{3P + \sigma_2^2 - \sigma_3^2}{2}$$

$$C = \frac{1}{2}\log\left(1 + \frac{P_2}{\sigma_2^2}\right) + \frac{1}{2}\log\left(1 + \frac{P_3}{\sigma_3^2}\right) \qquad \text{[2E]}$$

$$= \frac{1}{2}\log\left(1 + \frac{3P - \sigma_2^2 + \sigma_3^2}{2\sigma_2^2}\right) + \frac{1}{2}\log\left(1 + \frac{3P + \sigma_2^2 - \sigma_3^2}{2\sigma_3^2}\right)$$

iii)



Three channels is when

$$3P > 2\sigma_1^2 - \sigma_2^2 - \sigma_3^2$$

$$3P = \nu - \sigma_1^2 + \nu - \sigma_2^2 + \nu - \sigma_3^2$$

$$\implies \nu = \frac{3P + \sigma_1^2 + \sigma_2^2 + \sigma_3^2}{3} = P + \frac{\sigma_1^2 + \sigma_2^2 + \sigma_3^2}{3} \quad \text{[2E]}$$

$$P_1 = \nu - \sigma_1^2 = P + \frac{\sigma_2^2 + \sigma_3^2 - 2\sigma_1^2}{3}$$

$$P_2 = \nu - \sigma_2^2 = P + \frac{\sigma_1^2 + \sigma_3^2 - 2\sigma_2^2}{3} \quad \text{[2E]}$$

$$P_3 = \nu - \sigma_3^2 = P + \frac{\sigma_1^2 + \sigma_2^2 - 2\sigma_3^2}{3}$$

$$C = \frac{1}{2}\log\left(1 + \frac{P_1}{\sigma_1^2}\right) + \frac{1}{2}\log\left(1 + \frac{P_2}{\sigma_2^2}\right) + \frac{1}{2}\log\left(1 + \frac{P_3}{\sigma_3^2}\right) \quad \text{[2E]}$$

4.

a) i) We need to show that for FDMA

$$\left.\begin{aligned} R_1 &\le C\left(\frac{P_1}{N}\right) \\ R_2 &\le C\left(\frac{P_2}{N}\right) \\ R_1 + R_2 &\le C\left(\frac{P_1+P_2}{N}\right) \end{aligned}\right\} \text{CDMA region}$$

[2B]

where noise power $N = N_0 W$.

$$\frac{W_1}{2}\log\left(1+\frac{P_1}{N_0 W_1}\right) \le \frac{W}{2}\log\left(1+\frac{P_1}{N_0 W}\right) \qquad (1)$$

$$\frac{W_2}{2}\log\left(1+\frac{P_2}{N_0 W_2}\right) \le \frac{W}{2}\log\left(1+\frac{P_2}{N_0 W}\right) \qquad (2)$$

[2A]

$$\frac{W_1}{2}\log\left(1+\frac{P_1}{N_0 W_1}\right) + \frac{W_2}{2}\log\left(1+\frac{P_2}{N_0 W_2}\right) \le \frac{W}{2}\log\left(1+\frac{P_1+P_2}{N_0 W}\right) \qquad (3)$$

(1), (2) are true because

$$x\log(1+\tfrac{1}{x})$$

is increasing and

$$W \ge W_1, W \ge W_2 \ (W = W_1 + W_2) .$$

[1A]

(3) follows from concavity.

Now since

$$\frac{W}{P_1+P_2} = \frac{P_1}{P_1+P_2}\frac{W_1}{P_1} + \frac{P_2}{P_1+P_2}\frac{W_2}{P_2},$$

[1A]

we have

$$\frac{W}{2}\log\left(1+\frac{P_1+P_2}{N_0 W}\right)$$

$$= (P_1+P_2)\frac{W}{2(P_1+P_2)}\log\left(1+\frac{P_1+P_2}{N_0 W}\right)$$

$$\ge (P_1+P_2)\left[\frac{P_1}{P_1+P_2}\cdot\frac{W_1}{2P_1}\log\left(1+\frac{P_1}{N_0 W_1}\right) + \frac{P_2}{P_1+P_2}\cdot\frac{W_2}{2P_2}\log\left(1+\frac{P_2}{N_0 W_2}\right)\right]$$

$$= \frac{W_1}{2}\log\left(1+\frac{P_1}{N_0 W_1}\right) + \frac{W_2}{2}\log\left(1+\frac{P_2}{N_0 W_2}\right).$$

[2A]

[2A]

ii) The sum rate is given by

$$R = R_1 + R_2 = W_1 \log(1 + \frac{P_1}{N_0 W_1}) + (W - W_1) \log(1 + \frac{P_2}{N_0 (W - W_1)})$$

Let the derivative be 0: [3A]

$$\log(1 + \frac{P_1}{N_0 W_1}) - \frac{W_1}{1 + \frac{P_1}{N_0 W_1}} \frac{P_1}{N_0 W_1^2} - \log\left(1 + \frac{P_2}{N_0 (W - W_1)}\right) + \frac{W - W_1}{1 + \frac{P_2}{N_0 (W - W_1)}} \frac{P_2}{N_0 (W - W_1)^2} = 0$$

Solving this equation, we get

$$W_1 = \frac{P_1}{P_1 + P_2} W$$

[2A]

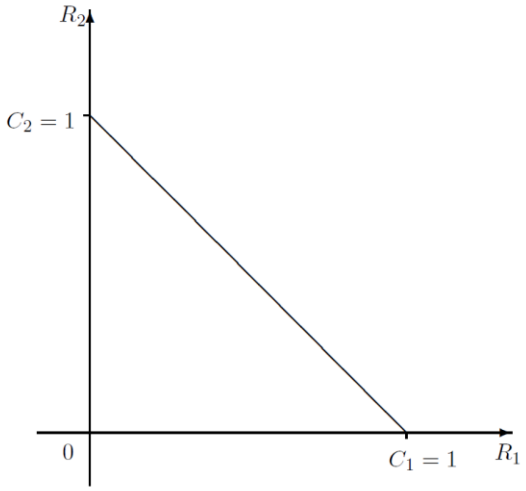
Alternative approach: students will also receive marks if concavity is used correctly to find the answer.

b)

i) $Y = X_1 \oplus X_2$. Clearly the total rate can't exceed 1 bit, since H(Y) = 1.

Sender 1 can achieve a rate of 1 bit by setting X2 = 0, and similarly Sender 2 can achieve 1 bit/transmission from. [3T]

By time sharing we can achieve the entire capacity region shown below.



[2T]

ii)

This channel is equivalent to a BEC (binary erasure channel) because

$$Y = \begin{cases} -2 & if\ X_1 = X_2 = -1 \\ 0 \quad if\ X_1 = 1, X_2 = -1\ or\ X_1 = -1, X_2 = 1 \\ 2 & if\ X_1 = X_2 = 1 \end{cases}$$

Hence the capacity region is

$$\begin{aligned} R_1 &< I(X_1; Y|X_2) = H(Y|X_2) \\ R_2 &< I(X_2; Y|X_1) = H(Y|X_1) \\ R_1 + R_2 &< I(X_1, X_2; Y) = H(Y) \end{aligned}$$

Choosing a uniform input distribution

$$P(X_1, X_2) = \frac{1}{4} \quad for\ (X_1, X_2) = (\pm 1, \pm 1)$$

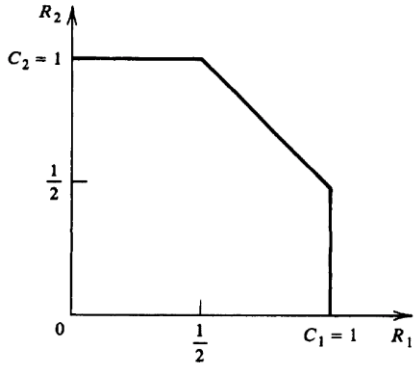
we have

$$H(Y|X_1) = H(Y|X_2) = 1, \qquad H(Y) = 1.5$$

Recall the capacity of BEC, the capacity region of the channel is

$$\begin{aligned} R_1 &< 1 \\ R_2 &< 1 \\ R_1 + R_2 &< 1.5 \end{aligned}$$

[3T]



Capacity region

[2T]